

Abstract

In Canada, prescription drugs and dental care have traditionally been excluded from the universally covered health services. In the absence of universal drug (and dental) insurance, low-income children have faced a higher risk of not receiving necessary treatment and their families faced the risk of high medical spending. In this paper I study the effects of the Quebec universal drug insurance on household spending. I estimate a small reduction in household drug spending among middle-income families but no effect for low-income families. Instead, I find larger offsetting increases in spending on health premiums for all income groups. This finding is alarming since it suggests an increasing financial burden from a drug insurance program instead of a benefit.

List of Abbreviations and Symbols Used

ACHB	Alberta Child Health Benefit
CRVE	cluster-robust variance estimator
FAMEX	Family Expenditures Survey
NCB	National Child Benefit
RMSPE	root mean squared prediction error
SCHIP	State Children's Health Insurance Program
SFHB	Saskatchewan Family Health Benefit
SHS	Survey of Household Spending
WCB	wild cluster bootstrap

0.1 Introduction

Providing health insurance for children has become a priority for many governments. This is reflected, for example, in the high policy profile of health insurance expansions for children in the US (Medicaid and the State Children’s Health Insurance Program (SCHIP)). Even though the morbidity and the costs of children’s health care are much smaller than those of adults, providing insurance for children is an important investment in their future (Currie, 2009; Goodman-Bacon, 2021). In Canada, where hospital and physician services have long been universally covered, national universal prescription drug coverage is in the early stages of development. In this paper, I study the effects of a universal drug insurance reform that was implemented in the province of Quebec in 1997. The reform introduced a new public drug plan for all uninsured families, provided free access for eligible children, and made prescription drug coverage mandatory in Quebec. The reform was followed by a sharp increase in prescription drug coverage and reportedly higher prescription drug utilization (Wang et al., 2015). While it was assumed to lower out-of-pocket costs (Furzer et al., 2023; Lebihan, 2023) and potentially reduce financial risks, no thorough estimations of these effects have been documented. In this study, using two different empirical strategies, I estimate a small reduction in household drug spending among middle-income families but find no evidence of statistically significant reductions in spending among low-income families. Instead, I find larger offsetting increases in spending on health premiums for all income groups, although only statistically significant effects for the middle-income group.

Traditionally, research on health insurance has been concerned with its effects on utilization and health outcomes (Currie & Gruber, 1996; Currie et al., 2014). More recently, research in the US has looked at the effects of insurance expansions on household spending and medical debt. The findings point to a substantial reduction in financial risk and medical and non-medical debt. These findings hold both in the general population (Finkelstein et al., 2012) and among low-income families with children, who were the main beneficiaries of the early Medicaid expansions (Gross &

Notowidigdo, 2011). These results are not surprising considering the high financial risk of having to face costs for all medical care, including physician and hospital services, in the US.

The existing Canadian literature on the burden of prescription drug costs is largely descriptive in nature (Caldbeck et al., 2015; McLeod et al., 2011). An exception to this is the study by Alan et al. (2005), who investigate the impacts of introducing public drug plans in Canada that cover extremely high costs, known as “catastrophic”¹ drug plans. Most of these plans were initiated in the 1970s, except for the Ontario Trillium program, which started in 1995. Looking at the budget shares of medication spending and focusing on the re-distribution effects, Alan et al. (2005) found larger reductions in spending for low-income families compared to higher-income families and concluded that these programs were strongly re-distributive.

In this paper, I investigate what effect the introduction of the Quebec public plan with zero copayment for children, and low copayment for parents (100 dollars annual deductible and a 25 percent co-insurance, in the first years of the program) had on out-of-pocket prescription drug expenditures of families with children. Considering that the public plan covered previously uninsured families and that higher-income families were more likely to hold (or be eligible for²) private insurance before the reform, I make a special emphasis on analyzing the heterogeneity of the effects by income group. Since the Quebec public drug plan was premium-based, I also analyze the effect of the reform on out-of-pocket spending on health premiums.

Using the fact that Quebec introduced a new mandatory public plan while other provinces did not, I use a quasi-natural experiment framework and compare the spending patterns of the treated (Quebec) and not treated (rest of Canada) families before and after the reform. In addition to the more traditional difference-in-differences method, I also apply the synthetic control method. Instead of relying on common trends assumption, the synthetic control method uses a linear combination of comparison units weighted to best match the pre-reform outcomes of the treatment group.

¹The high-deductible insurance plans which require a deductible on the order of 3-4 (or more) percent of family income.

²Anyone eligible for private insurance through their employer had to register for that private insurance rather than the public plan.

The resulting synthetic control group better approximates the treatment group than any single control unit. In addition, the method provides a reliable inference procedure and a clear graphical exposition.

I study the mean effects of the reform on household annual drug and health premium spending. Using the Statistics Canada Survey of Household Spending (SHS), for the Quebec universal plan, I find a mean reduction in annual drug spending of around \$45 (approximately 30 percent) for moderate-income households (\$30,000-\$50,000) but no effect for the lowest-income households (below \$30,000). The estimated mean increase in annual premiums was larger, \$110 (approximately 30 percent). As a result, the net effect was an increase in mean combined spending on prescription drugs and health premiums, of about \$62 (12 percent). Moreover, due to rising premiums, the net annual effects were increasing over time, reaching an estimated \$256 in 2001.

This paper contributes to the existing literature in several ways. First, it provides estimates of the effect of the Quebec drug insurance reform on household drug and health insurance spending, which has not yet been systematically studied. Understanding the scale of spending changes that the Quebec public program has brought about is crucial for shaping universal drug insurance policy design. The results suggest that in the first 5 years following the Quebec reform, mean household out-of-pocket spending on drugs reduced moderately, however, this reduction was more than offset by premium increases. Second, it finds important heterogeneity by income, with only the middle-income group benefiting from the reduction in out-of-pocket spending on drugs. Finally, it makes a methodological contribution. Because in the Canadian context of 10 provinces, the difference-in-differences method has to rely on a limited number of potential controls, the assumption of common trends may not hold. In this paper, I cross-validate the difference-in-differences results against those of the synthetic control method which requires fewer assumptions. The results suggest that the estimates for prescription drug spending and health premiums for the middle-income group remain robust to the choice of the method. Therefore, there is confidence that the results provide credible estimates of the effects of drug insurance on family spending.

The remainder of the paper is organized as follows. Section 0.2 lays out the details

of the drug insurance landscape in the Canadian provinces and the Quebec drug reform. Section 0.3 discusses the data sources and section 0.4 lays out the empirical strategy. The regression results are presented and discussed in Section 0.5. Section 0.6 concludes.

0.2 Drug insurance in Canada and the Quebec drug reform

This section provides the background on drug insurance in Canada and the Quebec drug insurance reform.³

In Canada, where hospital and physician services have traditionally been universally covered, prescription drugs have not. Instead, according to Hoskins et al. (2019), there have been numerous public plans (over a hundred) as well as a variety of private plans, many of which necessitate a copayment of some kind. Public coverage for medications has traditionally been available to welfare recipients and, with some variation in generosity depending on the province, to Canadian seniors (Grootendorst, 2002). The middle-class Canadians holding full-time jobs with benefits have traditionally been covered by work group plans (Applied Management et al., 2000), but their generosity may also have varied. The group most likely to have gone without coverage for prescription drugs in Canada are those employed without benefits (Angus Reid Institute, 2020; Applied Management et al., 2000; Grootendorst, 2002). As a result, in Canada, many families with children who are neither seniors nor on welfare have been facing the risk of out-of-pocket payments when filling prescriptions. According to a recent poll (Angus Reid Institute, 2020), between 20 (Ontario) and 37 (Manitoba) percent of respondents paid half or more of the prescription drug costs out-of-pocket. In addition, according to recent Canadian findings, families without adequate drug coverage were reported to reduce spending on food and other necessities (Law et al., 2018), and eventually resort to borrowing (Kolhatkar et al., 2018).

To ensure children do not experience financial barriers to prescription medications, over the last couple of decades, several Canadian provinces have introduced drug

³It should be noted that at the time of writing, the federal government has announced its commitment to implementing the national universal pharmacare. However, the program is still in the initial stages of implementation. The description below reflects the drug insurance landscape in effect at the time of writing.

benefits for children. First-dollar comprehensive plans began to emerge in the 1990s, starting with the Quebec drug reform in 1997 and several income-targeted programs in the late 90s in Alberta and Saskatchewan. More recently, Ontario introduced a comprehensive drug benefit for all uninsured children, Ontario Health Insurance Plan+ (OHIP+), in January 2018.

The Quebec universal drug plan is probably the oldest established program that has introduced universal drug coverage for children. The Quebec program was implemented in 1997, making drug insurance mandatory for all residents and introducing a compulsory public drug insurance plan for all uninsured residents. The plan charged income-indexed premiums, ranging between 0 and \$175, a monthly deductible of \$8.33 and a 25 percent copayment. However, the coverage was free for dependent children of enrolled (i.e., premium-paying) adults.

The design of the Quebec public plan is unique in the Canadian context. While there have been long-established universal public programs in several Canadian provinces (e.g., Saskatchewan, Manitoba, British Columbia) for “catastrophic” drug costs, they usually cover only the costs above a certain percentage of family income. Unlike those programs, a comprehensive plan such as the Quebec drug plan for children, removes all financial risks related to prescription drug spending, not only the risk of very high spending.

There were several other initiatives that provided children with new comprehensive drug benefits: Alberta Child Health Benefit (ACHB) (introduced in August 1998), and Saskatchewan Family Health Benefit (SFHB) (August 1998).⁴ The programs were provincial initiatives implemented as part of the National Child Benefit (NCB)⁵ re-investments and were narrowly targeted at low-income children. Table ?? in Appendix B summarizes the new public drug plans.

During the same period, Ontario introduced its catastrophic plan in 1995; and in 1996 Manitoba restructured the deductible schedule of its “catastrophic” plan, essentially

⁴In 1999, Prince Edward Island also introduced its Family Health Benefit, which I do not include in the list because it was much less “comprehensive”: it required a deductible for each prescription and a dispensing fee. Eligible were families with incomes below \$21,000.

⁵More details on the NCB are provided in Appendix A.

lowering the out-of-pocket cost for lower-income families and increasing it for higher-income families. A previous study reported little effect of the Ontario reform on drug spending budget shares (Alan et al., 2005).⁶ The available evidence for the effects of the Manitoba reform suggests that utilization did not increase for low-income children, while it decreased for higher-income children (Kozyrskyj et al., 2001). Considering that few families gained coverage through the Alberta and Saskatchewan children's drug programs and that the catastrophic drug plans in Manitoba and Ontario were unlikely to have any effect on mean drug spending, I keep these provinces in the main analysis.

0.3 Data

To analyze the effects of drug insurance on household spending, I use the SHS public-use microdata files by Statistics Canada. The SHS is a national cross-sectional survey which collects detailed information on household expenditures. It also contains information on income and its sources, geographic identifiers, and household demographic information. It is the only national source of health spending data for Canadian households, including prescription drug spending. Since the policy change I am interested in occurred in 1997, I combined the annual 1997-2001 cycles of the SHS survey with its predecessor, the Family Expenditures Survey (FAMEX), of which I use the 1992 and 1996 cycles.⁷ The major spending categories and demographic information are preserved in the FAMEX.

I restrict attention to respondents with children under the age of 18, living in one of the 10 provinces, who are under 65 years old and do not receive welfare. This is to ensure other public drug programs - such as those for seniors and welfare recipients - do not interfere with the analysis.

The key variable of interest is prescription drug spending. It has a long right tail, reaching up to \$25,000 in my sub-sample. I drop a small number of observations (55 observations - less than 0.2 percent of the sample) with drug spending above \$3,000.

⁶The authors suggest that low take-up and high deductibles could be the reason for the lack of any effect on household spending.

⁷Earlier cycles of FAMEX do not contain province identifiers but only geographic regions that combine multiple provinces.

Retaining these observations, which may have been a mistake in data entry or recall, could seriously impact regression results. The resulting data set spans the period between 1992-2001 and contains 28,341 observations.

A limitation of these data is that it is not known what part of total drug expenditure goes to children's drugs. The reform in Quebec covered eligible parents as well as children, but parents had to face copayment requirements. Therefore, part of the total change in drug spending could be the change in parents' drug spending. This, however, would only be true for the parents spending above the annual 100-dollar deductible.⁸

0.4 Empirical strategy

I am interested in studying the relationship between household drug expenditures and access to comprehensive coverage for prescription drugs among families with children. To estimate the effect of insurance on spending, I use an exogenous policy change - the introduction of mandatory public drug insurance coverage for all uninsured persons in Quebec starting in January 1997. Since no other province experienced a similar change in drug coverage, my general strategy is to compare the out-of-pocket drug expenditures before and after the reform in the treatment (Quebec) and control (other than Quebec) provinces.

The baseline difference-in-differences model is presented in Equation (1):

$$DRUG_{ipt} = \beta_0 + \beta_1 Que_p \times Post_t + \beta_2 Prov_p + \beta_3 Year_t + X'\Lambda + e_{ipt} \quad (1)$$

where the dependent variable, $DRUG_{ipt}$, is spending on prescription drugs by household i in province p and year t . $Que_p \times Post_t$, the interaction between Quebec and the year 1997 and after, is the key variable of interest. The coefficient β_1 captures the change in annual family spending on prescription drugs associated with obtaining comprehensive public drug insurance. $Prov_p$ and $Year_t$ are provinces and years fixed effects. X is a set of controls that include household size, age, sex and marital status of the reference person, and family income characteristics. All regressions use survey

⁸The mean of drug spending in 1996 was 142 dollars.

weights provided by Statistics Canada. Provinces and years fixed effects control for year-specific shocks common to all provinces (such as national programs, e.g., NCB) as well as province-specific time-invariant differences in drug spending levels which could be due to provincial public drug subsidy programs already in place (such as the Ontario Trillium Drug program).

My focus is on low- and middle-income households which were more likely to be affected by the public drug reform compared to higher-income groups in Quebec. Higher-income families were more likely to have been privately insured for drugs both before and after the Quebec reform.

Therefore, I estimate Equation (1) for low-, middle- and higher-income families separately. I define “low income” as incomes below \$30,000,⁹ “middle income” as incomes between \$30,000 to \$50,000 and higher incomes as incomes between \$50,000 to \$70,000. I set the low-income cut-off at \$30,000, adjusted for the Consumer Price Index (CPI) to ensure a reasonable sample size for the low-income group. This also allows me to align the study of the Quebec program with the Saskatchewan program studied in Chapter 4, where eligibility was linked to an income level below \$26,000. For higher income sub-samples, i.e. above \$50,000, I expect to see less of an effect as the drug insurance coverage, either public or private, did not increase as dramatically for this group as it did for the lower income groups.

Theoretically, health insurance does not need to result in lower health spending. First, non-adherence among the uninsured leads to lower utilization and spending on medications. Second, health-seeking and physician’s prescribing behaviour may be altered by the presence of insurance, i.e. lead to higher utilization, and, possibly, higher spending for the insured, if copayment requirements are present. Lastly, other categories of spending, more elastic to income, could have been sacrificed when the choice had to be made between spending on medication and other items, which would further downplay the effect of insurance on drug spending alone. There is some evidence that spending on children’s medication could be less price-elastic (Karaca-Mandic et al., 2012) and medication utilization is known to have lower levels of cost-related

⁹All expenditures are adjusted for inflation using provincial CPI with the base year 2000. Household income is not adjusted for family size, but all regressions control for household size.

non-adherence among children (Law et al., 2018), so the effect of comprehensive insurance for children’s drugs should be working towards reducing household out-of-pocket spending on medications. However, when, in addition to children, parents get coverage but must face copayments, the negative effect could be more muted.

The identifying assumption of the difference-in-differences design - the parallel trends - requires that the trends in out-of-pocket drug spending among families in the affected and the comparison provinces would have been the same in the absence of the reforms. There are reasons to believe this. There were no other major drug insurance changes during the same years.¹⁰

A potential threat to the validity of this difference-in-differences strategy is the simultaneous introduction of other programs during the same period that could have affected spending on prescription drugs. One prominent program launched during this period was the NCB. Introduced in 1998, it was a new supplement that provided cash benefits to families with children (more details on NCB are provided in Appendix A). I take advantage of the fact that most provinces implementing the NCB supplement including Ontario, British Columbia, and Manitoba, did not invest the funds into child drug benefit programs. And the three provinces that did (Alberta, Saskatchewan, and Prince Edward Island) imposed such low (see Table ??) income eligibility requirements, that very few non-welfare receiving families could qualify, and those families would all be included in the below \$30,000 income group. I conclude that there was little confounding effect of the NCB on trends in drug spending during this time.¹¹

For inference, it is common in the difference-in-differences literature to cluster standard errors at the level of the reform, i.e., at the province level. However, using the cluster-robust variance estimator (CRVE) may be unreliable with few (in this case just one) treated groups. It has been reported (MacKinnon & Webb, 2018) that with

¹⁰There were some changes to the catastrophic (last-dollar) plans: Ontario introduced its catastrophic plan in 1995 and Manitoba changed its catastrophic plan’s schedule of deductibles in 1996. The effects of these plans on mean spending are small, according to (Alan et al., 2005). If there were effects on spending, they would bias the estimates of the Quebec plan toward zero.

¹¹If there are some unaccounted effects of income transfers on drug spending, I expect them to work in the direction of increasing the spending by relaxing the income constraint and attenuating the negative effect of insurance.

few treated clusters, using a CRVE results in severe over-rejection with t-statistics being up to five times too large. Likewise, using wild cluster bootstrap (WCB) with very few clusters has been reported to perform poorly in simulations (MacKinnon & Webb, 2018) resulting in severe under-rejection (restricted bootstrap with null imposed) and under-rejection (unrestricted bootstrap). MacKinnon and Webb (2018) suggest that whenever these two wild bootstrap tests disagree, a “subcluster” test - a wild bootstrap clustered at a finer level (e.g., a simple wild bootstrap) - could provide a better solution, provided the clusters are of similar size. In this paper, I perform tests using both the restricted and unrestricted wild cluster and subcluster bootstraps. When they agree, I use them for inference. When they disagree, other inference approaches, such as the randomization inference (as is used with the synthetic control method), should be used. The alternative procedure that I use is discussed below.

In addition, although other provinces did not undergo major drug reforms, the potential for varying trends in drug spending across provinces and among different income sub-groups still exists. Such variations could violate the identifying assumption of the difference-in-differences analysis, necessitating further investigation.

To complement the difference-in-differences analysis, I estimate the effect of the Quebec drug reform using a synthetic control method. One of its major advantages is that it does not rely on parallel trends. Furthermore, it is designed for causal inference with one or a few treated groups, overcoming the limitations of the difference-in-differences inference with a small number of treated groups. The synthetic control method is an approach to program evaluation, introduced in Abadie and Gardeazabal (2003) and Abadie et al. (2010). It uses a weighting procedure to construct a synthetic comparison group from the existing potential control units which are chosen and assigned weights to best approximate the relevant pre-reform characteristics and outcomes of the treatment group. This proximity is based on a distance measure, root mean squared prediction error (RMSPE), applied to the pre-reform characteristics and outcomes. As a result, the comparison group comprising a linear combination of units weighted to minimize RMSPE provides a better approximation of the treated group than any single unit. In addition, it provides a transparent and

reliable placebo-based inference procedure. By constructing a synthetic control for each province as a placebo treatment, one obtains a distribution of treatment effects for which one can calculate a p-value. The p-value in this case is the proportion of placebos that have a ratio of post-treatment RMSPE (the gap between the observed and predicted outcomes) over pre-treatment RMSPE (which here also stands for the quality of the pre-treatment match) at least as large as the average ratio for the treated units. In other words, the probability of obtaining the observed difference in the outcomes by chance only. I compare this p-value to the p-values of the WCB and the subcluster bootstrap from the difference-in-differences models, and base inference on the former when the latter two p-values do not agree. Finally, having a single control group makes graphical exposition appealing. Therefore, following the literature, I present the graphs of trends, effects, and placebo effects, in addition to the estimates of the average treatment effects. In this context, I expect to see more extreme effects for the treated group (Quebec) compared to placebo effects among low-income groups, and less extreme - for higher-income households, considering that higher-income households have a higher probability of holding drug insurance before the reform.

0.5 Results

First, I present the summary statistics for the main outcomes and the covariates for three groups: Quebec (treated group), rest of Canada (non-treated group) and synthetic Quebec¹² over the pre-reform years. In my sample,¹³ households had less than 2 children on average, the average age of the reference person¹⁴ was between 37 (rest of Canada) and 38 (Quebec). The proportion of single-parent households was slightly higher in Quebec (11 percent) compared to rest of Canada (8 percent). The average income was lower in Quebec (\$49,130) than in rest of Canada (\$54,600). The synthetic Quebec demographic characteristics are almost identical to those in

¹²The demographic and prescription drug spending means in column 3 of Table 1 are based on the estimated synthetic control for prescription drugs using all income groups. The means of total insurance premiums in column 3 are based on the estimated synthetic control for total insurance premiums.

¹³My sample includes non-senior families with children not receiving social assistance.

¹⁴In the SHS, the “reference person” is defined as the household member who is “mainly responsible for its financial maintenance (e.g., pays the rent, mortgage, property taxes, and electricity)”.

rest of Canada and very similar to Quebec, except for the female reference person rate, which is much lower in Quebec between 1992-1996, but increases afterwards. Synthetic Quebec does not have a better match to Quebec in terms of demographic characteristics than rest of Canada. With only two pre-reform periods, I use both pre-reform observations on the outcomes and therefore matching is done only on the outcome variables and does not use the covariates.¹⁵

In terms of outcome variables, drug spending in Quebec was lower at the beginning of the period but showed some growth, unlike rest of Canada where it remained unchanged. Health premium spending in Quebec and rest of Canada was 3-4 times as large as drug spending, and in both cases showed some decline during the pre-reform years. For drug spending and total health insurance premiums, synthetic Quebec achieves a much better match on pre-reform trends than does the simple average of rest of Canada.

¹⁵According to Galiani and Quistorff (2017), unless some pretreatment outcome variables are dropped from the set of predictors, all other covariates are rendered redundant.

Table 1: Pre-reform characteristics and spending, 1992-1996

	Quebec	Rest of Canada	Synth Quebec
<i>Panel A. Demographics</i>			
Age of respondent	38.1	37.4	37.1
Female respondent	0.20	0.43	0.41
Single parent	0.11	0.08	0.08
Low education	0.46	0.43	0.43
N of children	1.67	1.75	1.75
Household income, 2000	49,130	54,599	53,576
N	910	4,431	.
<i>Panel B. Average health spending, \$</i>			
Prescription Drugs, 92	108.4	132.8	118.3
Prescription Drugs, 96	141.8	132.2	138.7
Total ins premiums, 92	415.4	431.4	415.3
Total ins premiums, 96	390.8	376.9	390.6

Notes: The sample includes families with children, not on welfare, where adults are less than 65 years old. Drug spending is limited to \$3000. All spending is adjusted for inflation using 2000 prices. N is not reported for synthetic Quebec as it is based on a weighted combination of aggregate data. The means of demographic variables and prescription drug spending for synthetic Quebec are based on the estimated synthetic control from the full-sample analysis of prescription drugs. Mean insurance premiums are based on the estimated synthetic control from the full-sample analysis of insurance spending.

Next, I examine the results of the difference-in-differences and the synthetic control analyses for mean drug spending. These results are presented in Table 2 and Table 3. Table 2 reports the results of the difference-in-differences estimation, with the results for different sub-samples in columns. In the first row, I show the estimated coefficient, and below it, I report several types of p-values. The first is based on CRVE, and it is followed by WCB estimators. The latter were clustered at different levels: *province* and *province* $\times$ *year* (i.e., the subcluster bootstrap, following MacKinnon and Webb (2018)). Acknowledging that with one treated group the CRVE as well as clustered bootstraps may be unreliable, I am looking for consistency between p-values within the same cluster level and, ideally, between the cluster levels. Using the whole sample of families with children, I do not find any effect on drug spending. However, the

results by income sub-sample suggest a reduction in spending among the middle-income group of about \$43, or 31 percent of pre-reform mean drug spending, but no effect for the other two income groups. The p-values, however, are not all consistent for the middle-income group. The CRVE p-value rejects the null, the WCB p-values clustered by province do not reject the null, and the WCB p-values clustered by province and year again reject the null. I further investigate these results using the synthetic control approach below.

In Figure 1, I present graphically the synthetic control results for the 3 sub-samples. Panels A, C, and E present the trends for Quebec and its synthetic control. Panel A suggests that a synthetic control group with a good match in the pre-reform period could not be generated for the lowest-income sample. On the contrary, the graphs in panels C and E demonstrate a good match in the pre-reform period for the higher-income sub-samples. To estimate the effect of the reform, I take the difference between spending in Quebec and its synthetic version after the reform. To assess the significance, I compare the magnitudes of the estimated effects for Quebec to the distribution of placebo treatment effects (i.e. the estimated effects for other provinces), taking into account the pre-reform match quality between the treated (or placebo-treated) province and its synthetic control. Panels B, D, and F plot these effects for Quebec (black) and for all the control provinces (grey). The effect for the low-income group is very pronounced, but the poor pre-reform match renders it less reliable. For the middle-income group, there is a noticeable divergence in spending trends after the reform (panel C) and a large negative effect compared to the placebo effects (panel D). Finally, for the high-income group, there is not any clear divergence in drug spending after the reform (panel E) and therefore no distinctive effect relative to the distribution of placebo effects (panel F).

Table 3 presents the results of the synthetic control procedure including the weights that it assigned to the provinces.¹⁶ The point estimates for the overall and the lowest-income samples (the first two columns) are larger than those reported in Table 2. However, the null hypothesis of zero effect cannot be rejected based on the p-values which are larger than 50 percent. These are not the standard p-values, however: they

¹⁶In each column, missing weights indicate that these provinces were dropped from estimation because synthetic control groups could not be generated for these provinces.

show the proportion of the control provinces that under placebo treatment scenario have an estimated effect at least as large as that for Quebec.¹⁷ Thus the p-values suggest a high probability (over 50 percent) of obtaining an effect similar to that in Quebec by chance only. Examining the synthetic control weights for these two samples (columns 1 and 2) reveals that only 2 out of 9 potential donors contributed in both cases, which would make it a sensitive and less robust control. Furthermore, as Panel A of Figure 1 demonstrates, there is a poor match in drug spending in pre-reform years. As a result, despite a pronounced negative effect observed in Quebec compared to the placebo effects (Panel B), the synthetic control p-values are too large. Conversely, for the middle-income group, there is a good pre-reform match between Quebec and its synthetic control and a distinctive gap in the after-reform years, with a maximum of \$63 in 1999. The average of these gaps gives the average treatment effect, which equals \$45 and is very close to the difference-in-differences estimate of \$42.6. The p-value is 0. Under the assumption of randomly assigned treatment, the true p-value could be calculated as the ratio of 1 (for Quebec) to 9 (all provinces), i.e. 0.11, which is relatively large but solely due to the small number of units in the analysis. Conversely, there is no sustained gap between Quebec and its synthetic version in the after-reform period for the higher-income group, hence the results suggest no effect of the reform for higher-income families.

The synthetic control analysis confirms the difference-in-differences analysis' finding that the Quebec drug reform led to a reduction of about 30 percent in drug spending among middle-income families. This is generally what one would expect - a reduction in spending among moderate- but not high-income households since the latter were more likely to have been privately insured for drugs before the reform. However, the results do not confirm that there was a reduction in spending for the lowest-income households either. The latter is likely due to the limited sample size of low-income families after I exclude all welfare-receiving households. A rather moderate reduction in out-of-pocket spending that I find for the middle-income group (despite the full coverage of children's drugs) is likely due to the presence of deductibles

¹⁷More specifically, to account for pre-reform match quality, these p-values calculate the proportion of control units that have a ratio of post-treatment RMSPE (the average gap) over pre-treatment RMSPE at least as large as the average ratio for the treated unit (Galiani & Quistorff, 2017).

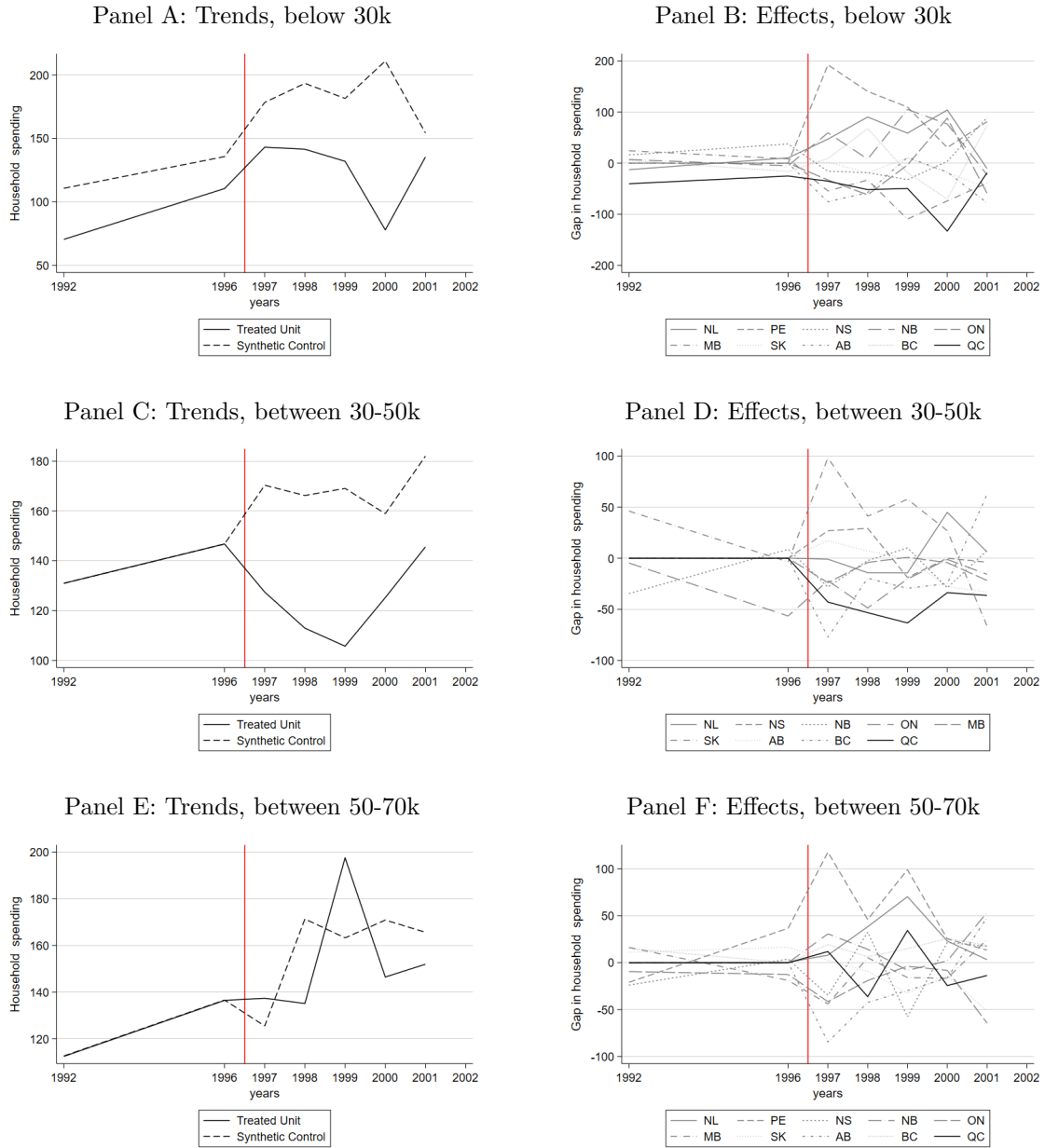
and copayments for adults. This finding suggests that, on average, only a small proportion of total drug budgets is spent on (commonly covered) children's drugs, and fully covering children's drugs has a small effect on the out-of-pocket spending of families with middle incomes.

Table 2: Effects of the Quebec drug reform on household average prescription drug spending, difference-in-differences

	All	below 30K	30/50K	50/70K
$Que \times Post$	-9.3	9.7	-42.6	2.9
p-value (CRVE)	0.215	0.426	0.005	0.804
p-val WCB (prov)	0.415	0.428	0.133	0.830
p-val restr WCB (prov)	0.508	0.482	0.380	0.830
p-val WCB (prov year)	0.645	0.814	0.050	0.908
p-val restr WCB (prov year)	0.707	0.814	0.090	0.909
N	28,341	4,576	10,763	7,788
Quebec N	3,845	678	1,617	1,011
Quebec pre-reform mean	122	88	139	123

Notes: This table reports estimates of the interaction of the treatment province and the post-reform years from the difference-in-differences model for prescription drug spending. Controls include year and province FE, family income, age, gender, female respondent, and family type. The estimation sample includes families with children, not on welfare, where adults are less than 65 years old. Drug spending is limited to \$3000. All spending is adjusted for inflation using 2000 prices. CRVE p-values are calculated using standard errors clustered by provinces. Additionally, I report the WCB p-values from restricted and unrestricted tests clustered at the province and province-year levels.

Figure 1: Prescription drug spending: trends and effects using synthetic control



Notes: The left-side panels show mean spending in Quebec (solid line) vs. its synthetic control (dashed line). The right-side panels show the gaps in spending for each province and its synthetic control. The vertical lines indicate the end of the pre-treatment period.

Table 3: Effects of the Quebec drug reform on household average prescription drug spending, synthetic control

	All	below 30K	30/50K	50/70K
<i>Panel A. Synthetic control effects</i>				
1997	-28.0	-35.2	-40.6	11.9
p-value	0.556	0.778	0.125	0.222
1998	-33.0	-51.9	-53.7	-36.2
p-value	0.556	0.778	0.000	0.222
1999	-35.4	-49.5	-63.4	34.3
p-value	0.556	0.889	0.000	0.222
2000	-47.1	-133.1	-30.9	-24.4
p-value	0.556	0.778	0.000	0.222
2001	-13.7	-18.8	-35.9	-13.7
p-value	1.000	1.000	0.125	0.222
ATT	-31.5	-57.7	-44.9	-5.6
p-value	0.556	0.889	0.000	0.222
<i>Panel B. Weights</i>				
NL	0.00	0.62	0.08	0.08
PE	0.36	0.00	.	0.06
NS	0.00	0.00	0.11	0.54
NB	0.00	0.00	0.26	0.08
ON	0.64	0.38	0.20	0.09
MB	0.00	0.00	0.11	0.03
SK	0.00	0.00	0.06	0.03
AB	0.00	0.00	0.10	0.06
BC	0.00	0.00	0.08	0.03

Notes: ATT is the average gap in the post-treatment years for Quebec. ATT, p-values, and synthetic weights are calculated using all non-missing provinces. Prince Edward Island is excluded from the synthetic control in column 3 because a synthetic control group could not be generated for this province in the placebo simulations.

Besides the effect on prescription drug spending, I am also interested in the effect of the Quebec reform on health insurance premiums. A public plan implemented in Quebec under the Universal Drug Insurance reform required families to pay income-indexed premiums. These premiums ranged from \$0 to \$175 in the first years but steadily increased reaching \$350 in 2000 and \$422 in 2002. It was argued that these premiums represented a serious financial burden for lower-income households, especially in the later years of the program when they were further increased (Morgan et al., 2017).

Table 4 reports the difference-in-differences estimates of the effect of the Quebec reform on total health insurance premiums. It suggests a consistent increase in spending, between \$117-\$136 for all income groups, unlike the effect on drug spending which I only find for the middle-income group. All p-values reject the null except the restricted CWB, which is known to underreject.

The synthetic control estimates in Table 5 also suggest a consistent pattern of positive effects for all income groups. In addition, unlike drug spending, the synthetic control weights for health premiums are more uniform and use the full pool of donors. The p-values, however, suggest that there is significant evidence of a positive effect only for the middle-income group, but not the lowest- or highest-income groups.

Figure 2 illustrates the synthetic control results. As Panels A, C, and D indicate, there is a good match in health insurance spending between Quebec and its synthetic control in pre-reform years (panels A, C, and D) and a noticeable divergence between them after the reform. The effects graphs (panels B, D, and F) show unusually large gaps for Quebec (dark line) in the after-reform years relative to placebo effects. There was also a sharp drop in premium spending in 1997, the year of the reform, for middle- and higher-income families. This drop is fully attributable to private plan premium spending and is likely due to the adjustment to the new policy among households and private insurers. Its large magnitude likely pushes downward the point estimates and inflates the p-values of the average post-treatment effects for the middle- and high-income groups. The years after 1997 show an unusually steep increasing trend in the effects on premium spending in Quebec relative to placebo effects for all income groups. While the average effect estimated using synthetic control is \$110, which

is a little under 30 percent of the pre-reform average insurance spending, the point estimate in 2001 suggests a much larger effect of \$287.7, which is above 60 percent of the pre-reform insurance spending.

To summarize, I find evidence of a substantial increase in health premiums, and a reduction, of a smaller magnitude, in drug spending for the middle-income group. The difference-in-differences and synthetic control estimates suggest premium increases for the other two income groups, but the large p-values from the synthetic control method indicate that these estimates are not statistically significant.

Since I find two opposite effects on spending for the middle-income group, I estimate the effect of the Quebec reform on the sum of drug and health insurance spending. The results are presented in Tables 6 and 7. The difference-in-differences estimates for the sum of the two categories in Table 6 are almost identical to the estimates for health premiums in Table 4 except for the middle group. The synthetic control estimate of the combined spending in Table 7 is also very close to the estimate for premium spending alone for the high-income group in Table 5. Similar to previous findings, the synthetic control p-values do not provide evidence of significant effects for the low and high-income groups but they do provide significant evidence of an *increase* in the combined spending for the middle-income group. Finding an increase in spending suggests a larger role of premium increases compared to the savings from out-of-pocket spending on prescription drugs. As is clear from panel D of Figure 3, the average effect is influenced by a drop in 1997, but starting in 1998 there is a pattern of steadily rising increases reaching up to \$256 in 2001.

The results suggest that the Quebec reform affected the out-of-pocket spending of the middle-income group by cutting drug spending but also increasing spending on health premiums. The combined average effect was a moderate increase in spending, of about \$62, or 12 percent, of pre-reform average spending. However, the pattern of the estimates indicates increasing positive effects on spending over time reaching up to \$256 in 2001. There is some indication of increased spending for the high-income group, attributable to higher premium spending, but I do not obtain consistent statistical evidence to support this finding.

Although my findings are specific to prescription drug coverage and the particular features of the Quebec public insurance plan (such as the premiums and the co-payment rates), the reduction in out-of-pocket health spending that I estimate is generally in line with the effects of public health insurance on spending reported in the literature. For example, programs such as Medicare (seniors), Medicaid (very low-income households) and SCHIP (State Children's Health Insurance Program for children in low-middle income families) have all been reported to reduce out-of-pocket health spending (Banthin & Selden, 2003; Finkelstein & McKnight, 2008; Leininger et al., 2010). While Medicaid and SCHIP target only low-income families, where I did not find a significant effect, Medicare - a premium-based public program covering seniors of all income levels - more closely resembles the design of the Quebec plan for middle-income households. Medicare was found to reduce out-of-pocket medical spending by between one-quarter (hospitals, not statistically significant) and one-third (physicians), and a more recent Medicare Part D program¹⁸ was found to reduce out-of-pocket drug spending by 13-18 percent (Polinski et al., 2010), whereas I find a one-third reduction in out-of-pocket drug spending for the Quebec universal public drug program. Medicare (not including Part D) was also found to increase total premium spending which at the mean outweighed the out-of-pocket health spending savings and led to an increase in net spending of about one-third (Finkelstein & McKnight, 2008), whereas I find a smaller net increase of about 12 percent on average, but getting much more pronounced in later years. The trend of increasing health spending due to rising premiums is alarming, as it suggests a potentially increasing financial burden instead of a benefit from the program, especially for lower-income families.

¹⁸Medicare itself does not cover prescription drugs. Medicare Part D, introduced in 2004, is provided by private insurance and the copayment requirements vary by plan.

Table 4: Effects of the Quebec drug reform on household average health insurance premiums, difference-in-differences

	All	below 30K	30/50K	50/70K
$Que \times Post$	122.2	136.5	117.4	135.9
p-value (CRVE)	0.000	0.001	0.000	0.000
p-val WCB (prov)	0.000	0.023	0.000	0.000
p-val restr WCB (prov)	0.202	0.148	0.132	0.173
p-val WCB (prov year)	0.194	0.081	0.177	0.075
p-val restr WCB (prov year)	0.199	0.161	0.188	0.132
N	28,341	4,576	10,763	7,788
Quebec N	3,845	678	1,617	1,011
Quebec pre-reform mean	412	159	374	506

Notes: This table reports estimates of the interaction of the treatment province and the post-reform years from the difference-in-differences model for health premium spending. Controls include year and province FE, family income, age, gender, female respondent, and family type. The estimation sample includes families with children, not on welfare, where adults are less than 65 years old. All spending is adjusted for inflation using 2000 prices. CRVE p-values are calculated using standard errors clustered by provinces. Additionally, I report the WCB p-values from restricted and unrestricted tests clustered at the province and province-year levels.

Table 5: Effect of the Quebec drug reform on household average health insurance premiums, synthetic control

	All	below 30K	30/50K	50/70K
<i>Panel A. Synthetic control effects</i>				
1997	-107.5	27.5	-134.6	-108.6
p-value	0.000	0.250	0.000	0.250
1998	76.6	76.6	102.0	117.2
p-value	0.000	0.250	0.000	0.125
1999	31.4	163.5	125.5	146.1
p-value	0.000	0.125	0.000	0.000
2000	145.5	238.6	172.5	155.6
p-value	0.000	0.250	0.000	0.125
2001	307.0	226.6	287.7	185.8
p-value	0.000	0.125	0.000	0.125
ATT	90.6	146.6	110.6	99.2
p-value	0.000	0.250	0.000	0.125
<i>Panel B. Weights</i>				
NL	0.03	0.02	0.11	0.09
PE	0.03	0.03	0.20	0.07
NS	0.19	0.04	0.15	0.15
NB	0.04	0.38	0.12	0.10
ON	0.03	0.47	0.22	0.06
MB	0.45	0.00	0.11	0.23
SK	0.03	0.06	.	0.07
AB	.	0.01	.	.
BC	0.21	.	0.09	0.23

Notes: Panel A reports the yearly estimated effects and the ATT which is the average effect in the posttreatment years for Quebec. ATT, p-values, and synthetic weights are calculated using all non-missing provinces. Panel B reports the synthetic control weights. A missing weight indicates that the corresponding province was dropped from the estimation because a synthetic control group could not be generated for this province in the placebo simulations.

Figure 2: Total health insurance premium spending: trends and effects using synthetic control

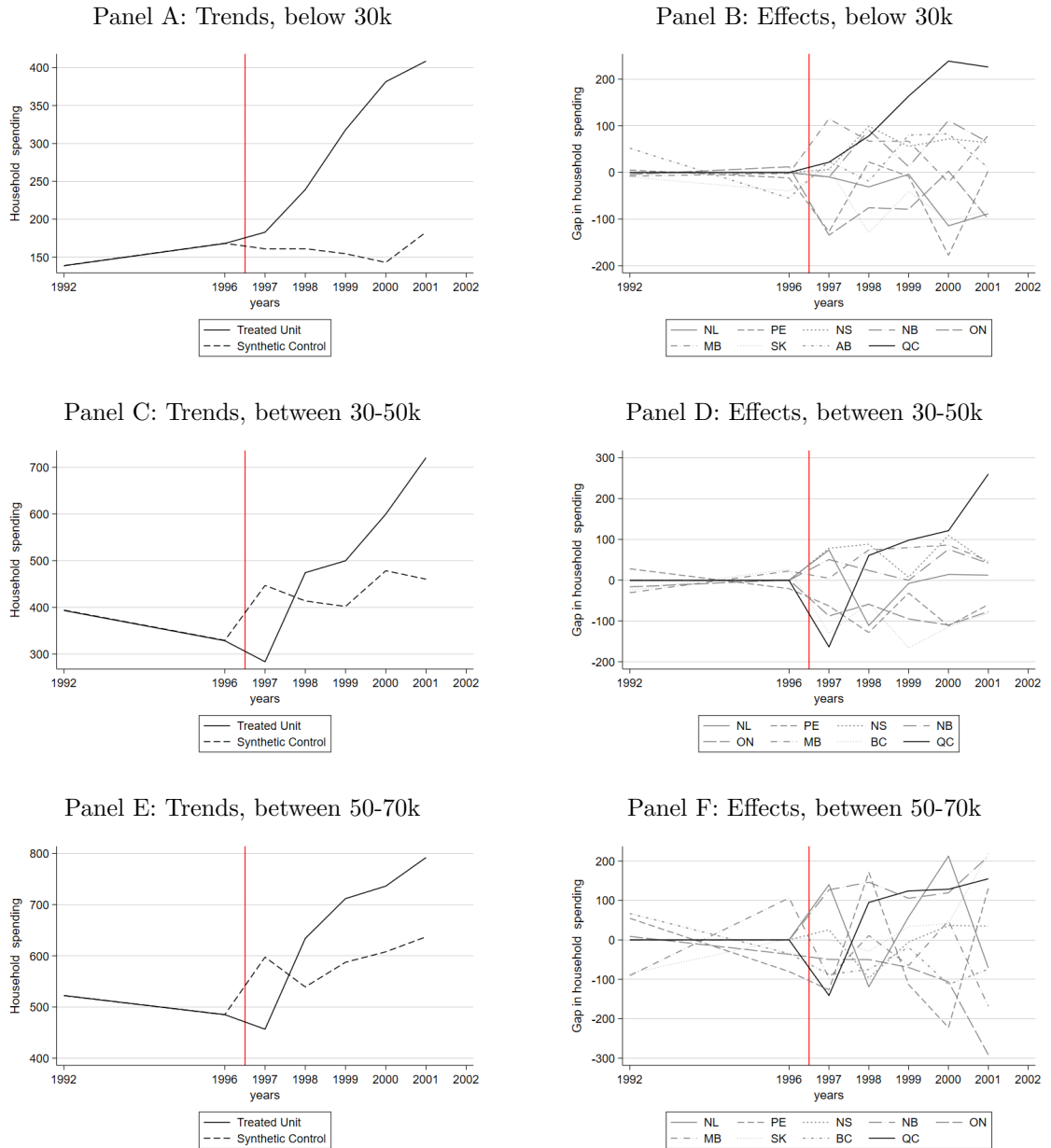


Table 6: Effects of the Quebec drug reform on combined prescription drug and insurance premium spending, difference-in-differences

	All	below 30K	30/50K	50/70K
$Que \times Post$	112.9	146.3	74.8	138.8
p-value (CRVE)	0.000	0.001	0.015	0.000
p-val WCB (prov)	0.000	0.000	0.001	0.000
p-val restr WCB (prov)	0.153	0.118	0.282	0.088
p-val WCB (prov year)	0.240	0.214	0.352	0.064
p-val restr WCB (prov year)	0.250	0.223	0.370	0.122
N	28,341	4,576	10,763	7,788
Quebec N	3,845	678	1,617	1,011
Quebec pre-reform mean	534	247	512	628

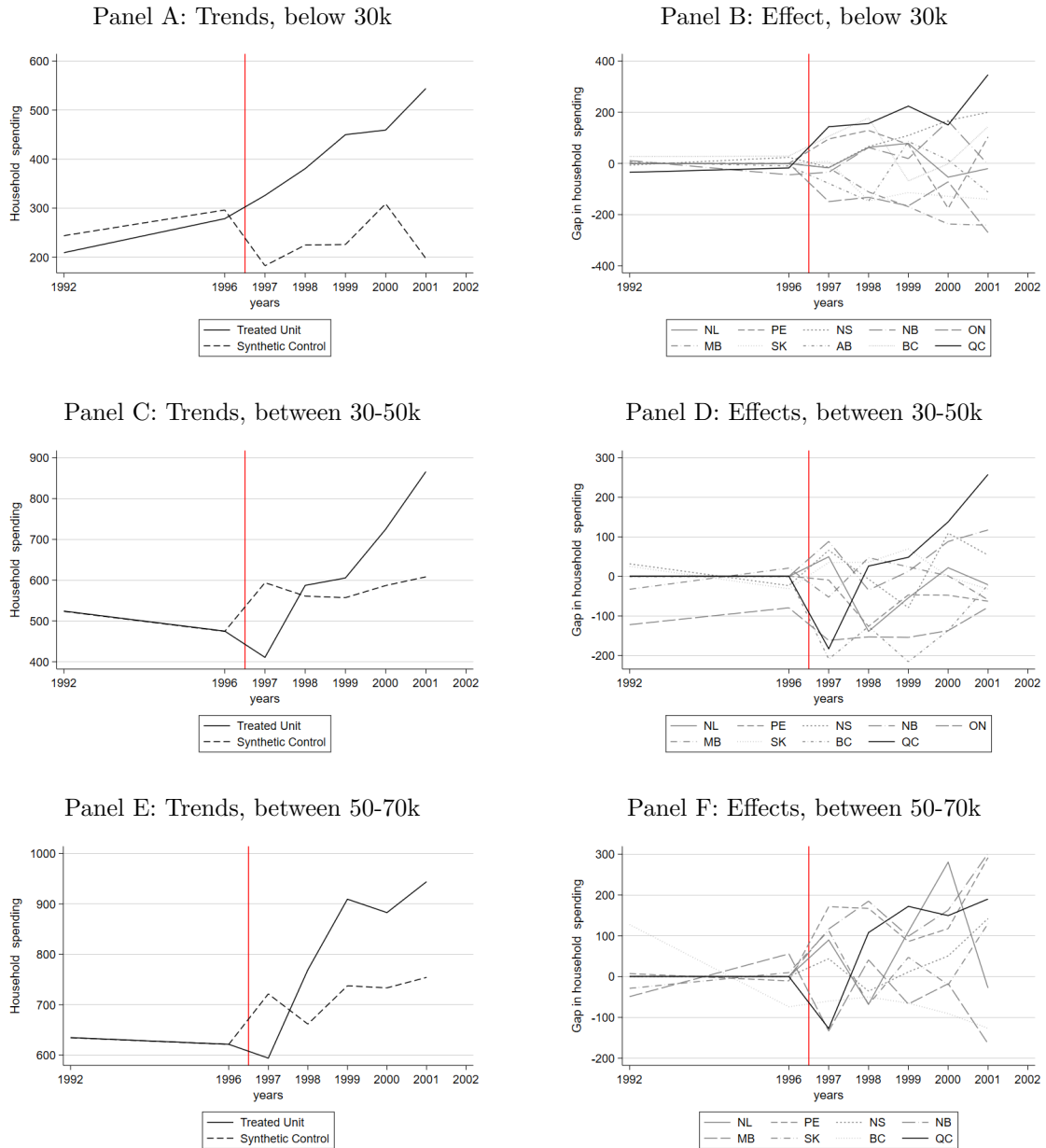
Notes: This table reports estimates of the interaction of the treatment province and the post-reform years from the difference-in-differences model for drug and health premium spending combined. Controls include year and province FE, family income, age, gender, female respondent, and family type. The estimation sample includes families with children, not on welfare, where adults are less than 65 years old. Drug spending is limited to \$3000. All spending is adjusted for inflation using 2000 prices. CRVE p-values are calculated using standard errors clustered by provinces. Additionally, I report the WCB p-values from restricted and unrestricted tests clustered at the province and province-year levels.

Table 7: Effects of the Quebec drug reform on combined prescription drug and insurance premium spending, synthetic control

	All	below 30K	30/50K	50/70K
<i>Panel A. Synthetic control effects</i>				
1997	-136.4	143.5	-178.8	-118.2
p-value	0.000	0.667	0.000	0.143
1998	44.9	156.0	36.1	87.7
p-value	0.111	0.667	0.125	0.143
1999	-2.1	224.2	59.6	175.4
p-value	0.333	0.444	0.000	0.000
2000	117.3	150.5	136.2	118.7
p-value	0.000	0.556	0.000	0.143
2001	309.0	346.9	256.3	182.8
p-value	0.000	0.556	0.000	0.286
ATT	66.5	204.2	61.9	89.3
p-value	0.000	0.556	0.000	0.286
<i>Panel B. Weights</i>				
NL	0.04	0.00	0.06	0.11
PE	0.05	0.00	0.07	0.10
NS	0.08	0.00	0.33	0.27
NB	0.05	0.00	0.12	0.12
ON	0.01	1.00	0.05	.
MB	0.58	0.00	0.28	0.23
SK	0.04	0.00	0.03	0.09
AB	0.11	0.00	.	.
BC	0.04	0.00	0.06	0.09

Notes: Panel A reports the yearly estimated effects and the ATT which is the average effect in the posttreatment years for Quebec. ATT, p-values, and synthetic weights are calculated using all non-missing provinces. Panel B reports the synthetic control weights. A missing weight indicates that the corresponding province was dropped from the estimation because a synthetic control group could not be generated for this province in the placebo simulations.

Figure 3: Sum of prescription drug and health insurance premium spending: trends and effects using synthetic control



Notes: The left-side panels show mean spending in Quebec (solid line) vs. its synthetic control (dashed line). The right-side panels show the gaps in spending for each province and its synthetic control. The vertical lines indicate the end of the pre-treatment period.

0.6 Conclusion

In this paper, I have examined the effects of the Quebec drug reform of 1997. The reform, uniquely among the Canadian provinces, made prescription drug coverage mandatory in the province of Quebec; and it introduced a new public drug plan for all uninsured families. The new public plan was more comprehensive than most other provincial plans available to working families at the time, which usually only covered very high, i.e., “catastrophic”, out-of-pocket spending. While there were studies of the effects on household spending of these “catastrophic” plans, there has not been a thorough investigation of the effects of the more comprehensive premium-based Quebec drug plan on household spending in the literature.

Using difference-in-differences and synthetic control approaches, I estimate the effects of the reform on mean household spending on prescription drugs and health premiums. I find a reduction in mean spending for prescription drugs after the Quebec reform of about \$45 (or 30 percent of the pre-reform average spending) among the moderate-income, but no strong evidence of an effect for the lowest-income households. This is despite the lower pre-reform rates of insurance coverage for the latter. The estimated effect on drug spending for the lowest-income group is an increase of \$9.7 using difference-in-differences, and a reduction of \$57.7 using synthetic control. Due to the limited sample size after excluding welfare recipients and the large variance, both approaches suffer from a poor match between the treated and control groups and are not statistically significant. Further studies using larger datasets, such as administrative data, are warranted to investigate the effect of free drug insurance on mean out-of-pocket spending among lowest-income working families.

The mean estimated dollar reduction in spending for prescription drugs, however, was largely offset by an estimated dollar increase in premiums spending: with both public and private plans charging premiums, all types of families faced a substantial increase in health premium spending. The estimates, however, were only significant for the middle-income group and suggested an increase on average between 1997 and 2001 of about \$110.6, or 30 percent, using the synthetic control method, and \$117 - using difference-in-differences.

This paper extends the literature on the Canadian healthcare system evaluations in several ways. First, it provides for the first time to our knowledge a thorough analysis of the effect of the Quebec drug reform on drug spending of families with children using quasi-experimental designs. Second, it applies the synthetic control method in the context of a few Canadian provinces. This approach is particularly suited for the case of a single treated and few control units, where the validity of the parallel trends assumption and the inferential procedure of the difference-in-differences are questionable. Finally, the results suggest that given the public plan’s copayment requirements, the reform did not provide a particularly large spending reduction for prescription drugs and led to a net increase of about 12 percent on average when premium spending was added. Moreover, the spending increases were getting much more pronounced in later years.

While the estimated reduction in mean spending following the reform is modest, it reflects full coverage of children’s costs, which constitute a small proportion of total household spending. The effect of covering adults is not captured at the mean since adults were required to pay deductibles equal to \$100 initially, which is comparable to the average pre-reform out-of-pocket drug spending of \$122, in addition to a 25 percent co-insurance. Families with above-average spending likely saw larger savings, but this is beyond the scope of the current study. Full coverage of children’s costs and a moderate deductible and co-insurance for adults under the Quebec plan compares favourably to the alternative, high-deductible (“catastrophic”) drug plan design that typically requires a deductible of around 3 percent of income, which would amount to \$750 for a family with an income of \$25,000. According to the Survey of Household Spending, very few families with children (less than 5 percent in my sample) spent that much on prescription drugs out-of-pocket, and with 3-percent income deductibles, many fewer families would benefit from the program. For comparison, the introduction of the Ontario Trillium plan, an income-indexed high-deductible universal program, did not affect mean drug spending shares (Alan et al., 2005). In addition, under a “catastrophic” plan, utilization may be reduced and financial strain may persist, compared to a more comprehensive plan like the one adopted by Quebec.

In addition, the results point to the trend of increasing health spending due to rising

premiums, which is alarming. This paper looked at the first five years of the Quebec reform when premiums were set between \$0 and \$175 in the first three years and increased up to \$350 in the next two. However, premiums continued to increase in the following years, reaching up to \$600 by 2010. As my estimates show, mean drug spending reductions were just a fraction of the premium increases, and the increasing spending on premiums after the reform likely represented an increasing financial burden for lower-income families. Thus, for a public program such as national universal pharmacare to reduce the financial burden of prescription drugs on families with children, the policy recommendation would be to limit copayment requirements, particularly for lower-income families. While the Quebec plan may have benefited financially certain groups of prescription drug users (such as very high spenders), the focus in this paper has been on families of low- and moderate incomes with average spending on prescription drugs. I find that they did not gain financially from this reform and would benefit from lower user charges.

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